System Capacity & Care Load Analytics for Unaccompanied Children

Exploratory analysis of CBP custody, transfers, HHS care and discharge reporting (2023–2025)

Abstract

This project develops a reproducible analytical framework for the reported care pipeline. The dataset contains 720 dated records from 2023-01-12 through 2025-12-21. The mean combined reported load was 6,233, with a maximum of 11,762 on 2023-12-20. Because dates are irregular and facility capacity is not included, results are descriptive and do not prove overcrowding or care quality.

1. Background and objectives

The CBP-to-HHS pathway can be studied as a dynamic operational pipeline. A useful monitoring framework distinguishes stocks (children reported in custody or care at a point in time) from flows (apprehensions, transfers and discharges over a reporting interval). This project summarizes reported load, compares CBP and HHS patterns, describes transfer/discharge balance, and supplies an interactive dashboard.

2. Dataset and preparation

Blank rows were removed, dates parsed, numeric values with comma separators converted, and records sorted. The dataset contains 720 records, 720 unique reporting dates, and 0 duplicate dates. It spans 1075 calendar days, leaving 355 dates not represented in the full calendar. Missing dates were not filled with fabricated values.

3. Analytical definitions

• **Total system load:** reported children in CBP custody + reported children in HHS care.
• **Net HHS intake (simplified):** transfers out of CBP custody – children discharged from HHS care.
• **Rolling average:** trailing seven-observation mean; irregular reporting means this is not necessarily a seven-calendar-day average.
• **Pressure signal:** repeated positive net-intake observations can flag a period for review but are not a direct measure of backlog or capacity.

4. Descriptive findings

• Mean CBP custody was 171; mean HHS care was 6,061.

• Mean combined reported load was 6,233; peak was 11,762 on 2023-12-20.

• Mean transfers out of CBP were 128.7 and mean HHS discharges were 173.4 per reporting record.

• Mean simplified net HHS intake was -44.7; net intake was positive on 33.1% of reporting records. This does not necessarily equal the change in HHS census.

• The reporting calendar is incomplete and irregular; comparisons should distinguish reporting-record averages from calendar-day estimates.

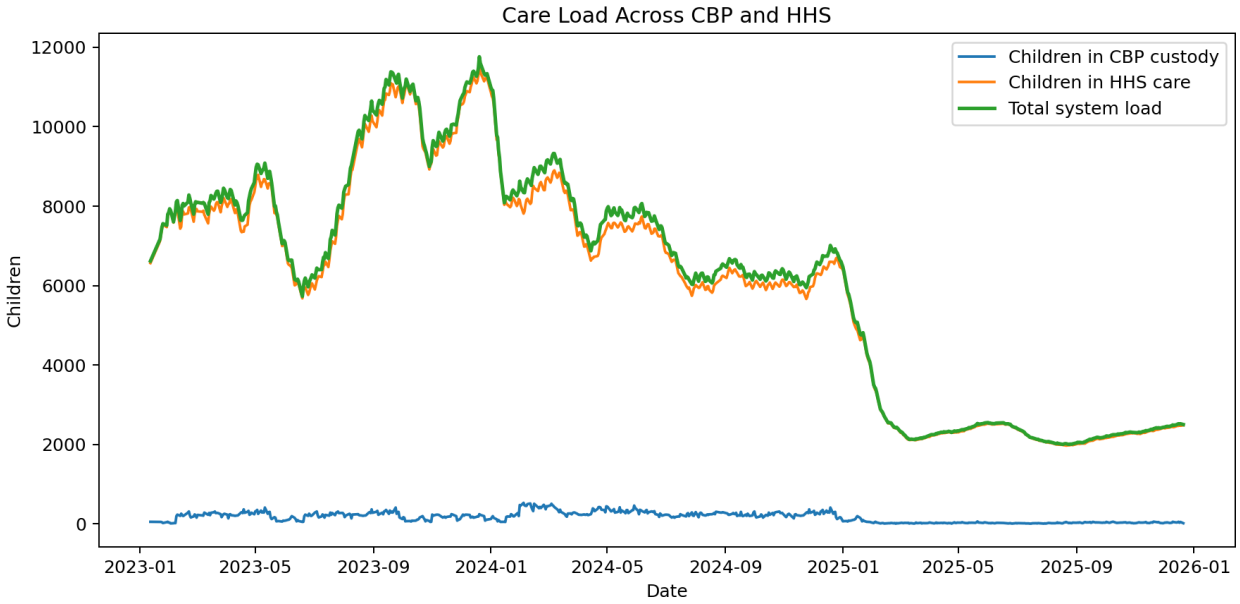


Figure 1. Reported CBP, HHS and combined care load.

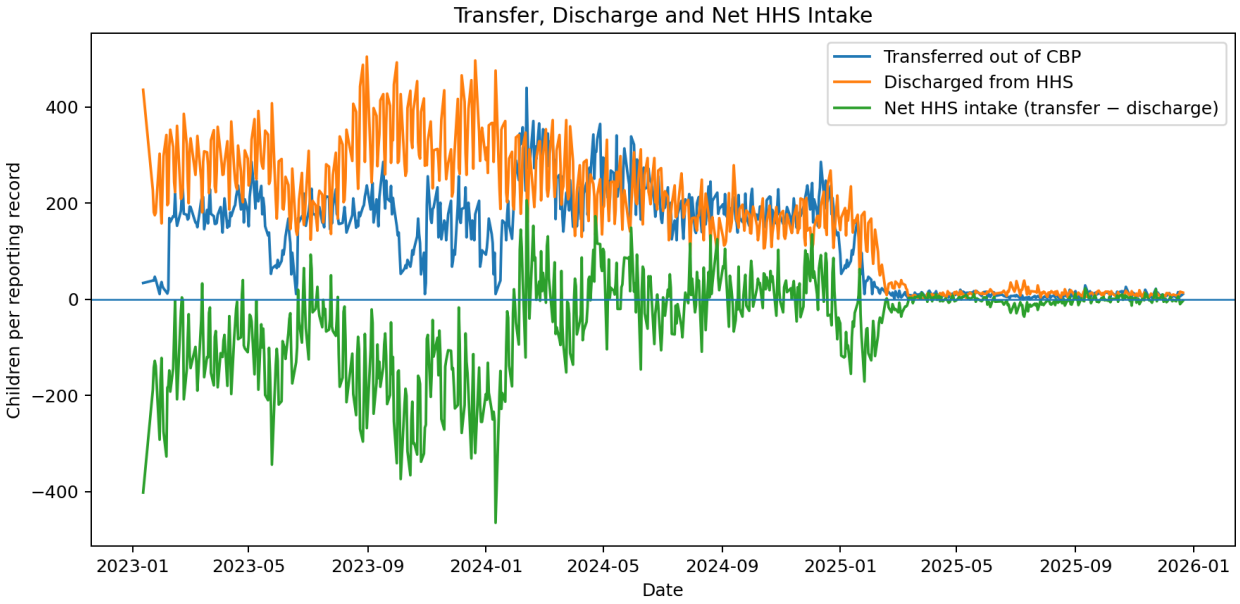


Figure 2. Transfers, discharges and simplified net HHS intake.

5. Dashboard design

The Streamlit dashboard provides a date-range selector, metric toggles, daily/weekly/monthly views, KPI cards, transfer/discharge charts, a net-intake view, data-quality notes, tabular results and CSV export. Resampled weekly/monthly values average the observed records in each period and may not represent complete calendar-period daily averages.

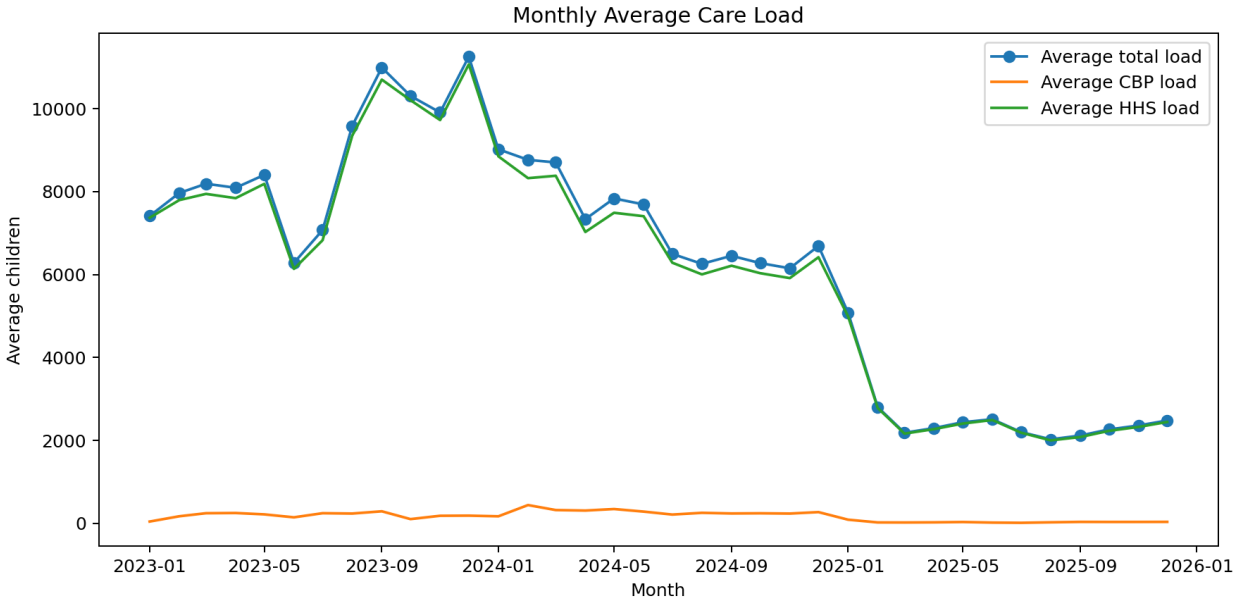


Figure 3. Monthly averages from available observations.

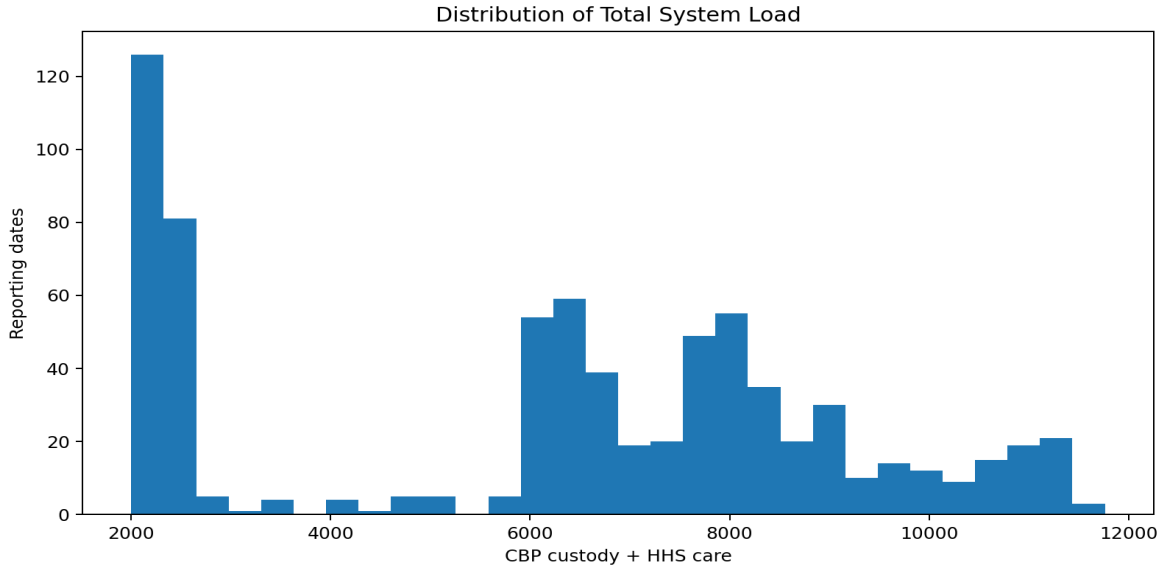


Figure 4. Distribution of combined reported load.

6. Recommendations

• Publish a consistent reporting calendar and document missing records; never silently treat missing dates as zero.

• Add facility capacity, staffed-bed capacity, occupancy, staffing, acuity, length-of-stay and placement-wait measures before asserting capacity stress.

• Set operational alerts only after thresholds and data-quality rules are validated.

• Separate stock and flow measures and display definitions next to each KPI.

• Display data freshness, missing-date counts and revisions prominently.

• Validate results with official source documentation and subject-matter experts before operational use.

7. Limitations and ethics

The data are aggregate and do not include individual-level records, formal capacity denominators, staffing adequacy, clinical need or case complexity. Combined CBP/HHS counts may not represent unique children if definitions or timestamps overlap. The simplified net-intake indicator is not an accounting identity for the HHS census. Do not use these results to make individual placement decisions or to infer misconduct.

Conclusion

This project offers a transparent baseline for monitoring reported load and flow balance. Its most defensible use is descriptive situational awareness, paired with improved reporting completeness and capacity data. A validated operational system requires official metric definitions, complete time coverage, capacity denominators and stakeholder review.

Data source: user-provided HHS_Unaccompanied_Alien_Children_Program.csv. Verify against official documentation before formal publication.